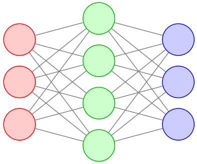
Theoretical Background: PPO Optimization

Actor Network Update



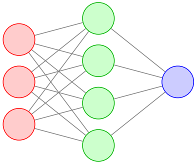
Maximize PPO Clipped Objective

$$\theta \leftarrow \theta + \alpha \nabla_\theta \mathcal{L}^{\text{CLIP}}(\theta)$$

$$r_t(\theta) = \frac{\pi_\theta(a_t^i | o_t^i)}{\pi_{\theta_{\text{old}}}(a_t^i | o_t^i)}$$

$$\mathcal{L}^{\text{CLIP}}(\theta) = \mathbb{E}\Big[\min\Big(r_t(\theta)\hat{A}_t, \text{clip}\big(r_t(\theta), 1-\epsilon, 1+\epsilon\big)\hat{A}_t\Big)\Big]$$

Critic Network Update



Minimize Value Loss (MSE)

$$\phi \leftarrow \phi - \beta \nabla_\phi \mathcal{L}^{\text{VF}}(\phi)$$

$$V_t^{\text{target}} = R^{(t)} + \gamma V_{\phi_{\text{old}}}(o_{t+1}^i)$$

$$\mathcal{L}^{\text{VF}}(\phi) = \mathbb{E}\Big[\big(V_\phi(o_t^i) - V_t^{\text{target}}\big)^2\Big]$$

$$\hat{A}_t = V_t^{\text{target}} - V_{\phi_{\text{old}}}(o_t^i)$$